Forecast the chute. Inform the operator.

Deep MLP forecasting for sugarcane feeding

PROPOSED RESEARCH



Mill context • supplied reference photo

INPUTS

Carrier signals
RPM • Current • Voltage

Steam flow
Shredder steam flow

Manual level estimates
Chute level • Observation age

History
Recent signals and estimates

→

MLP
Deep neural network

→

30 s forecast
Future chute level

→

Rule-based advice
Operational logic

→

Operator
Reviews and decides

SIMULATION / ILLUSTRATIVE

Current level
68%

Predicted level +30 s
72%

Target band
60 – 75%

Recommended load / feed
HOLD

Feed advisory • not a measured tonnage

Chute level forecast (simulated)





SIMULATION-FIRST

Baselines
Compare with simple forecasting methods

Feature ablations
Assess input contributions

Field validation planned
On-site evaluation in future work

Syed Affan Ali | Python • PyTorch • Time-series modeling

Concept visuals and sample values. Evaluation pending.